# Constraint-Aware Hierarchical Risk Parity for Robust Portfolio Allocation

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**Abstract**

Hierarchical Risk Parity (HRP) is a popular portfolio construction method that allocates capital through a correlation tree instead of inverting a covariance matrix, which makes it robust to estimation error. However, HRP cannot respect the constraints that real mandates impose: per-asset limits, sector or asset-class caps and floors, and turnover budgets. We propose Constraint-Aware HRP (CA-HRP), which keeps HRP's hierarchical risk allocation as a target and maps it to the feasible portfolio with minimum ex-ante tracking error to it. We prove that this projection is unique, leaves compliant HRP portfolios unchanged, and redistributes excess weight as a covariance-based hedge. We also prove that a solver-free capacity-aware bisection always meets per-asset bounds exactly, and we study a regime-robust projection with a worst-case guarantee over a convex set of covariance matrices. The design is fixed on a validation period and then tested on a held-out period, on three universes added only after the method was frozen (Indian ETFs, US sectors and global country funds), at three rebalancing frequencies, and in a Monte Carlo study with a known covariance — six universes and more than 1,300 monthly rebalances from 2001 to 2026 in total. CA-HRP satisfies every constraint at every rebalance, whereas HRP breaches them at 48–100% of rebalances and the common clip-and-renormalise fix at up to 46%. Where constraints bind hardest, CA-HRP raises the Sharpe ratio from 0.76 to 0.91 relative to Euclidean projection, with 25% lower volatility; it has the lowest or equal-lowest volatility of the HRP fixes in all six universes and the highest true Sharpe ratio of all methods in the simulation. It does not, however, beat constrained minimum variance on realised returns, and the regime-robust variant did not help in the 2022 stock–bond correlation shift. Most gains come from group constraints, and the risk-aware projection metric, not the estimator or clustering rule, drives them.

**Keywords:** hierarchical risk parity, portfolio optimisation, constrained allocation, tracking error, robust estimation, convex optimisation.

## 1 Introduction

Mean–variance optimisation [9] is the textbook answer to portfolio construction, yet in practice its outputs are notoriously unstable: small errors in estimated means and covariances are amplified by the inversion of an ill-conditioned covariance matrix into extreme, concentrated and rapidly changing weights [1, 10]. Out of sample, optimised portfolios frequently fail to beat naive equal weighting [2]. Two families of remedies have emerged. The first regularises the inputs, for example by shrinking the covariance matrix [5] or by imposing weight constraints, which Jagannathan and Ma [4] show act as a form of shrinkage. The second abandons optimisation altogether in favour of heuristic risk allocation, such as equal risk contribution [8].

Hierarchical Risk Parity (HRP), introduced by López de Prado [7], belongs to the second family and has become one of the most popular allocation methods among practitioners and in open-source libraries. HRP clusters assets by correlation, orders them so that similar assets are adjacent, and allocates capital top-down by recursively splitting each cluster in inverse proportion

to its risk. Because it never inverts the covariance matrix, HRP is robust to estimation noise and can handle singular covariance matrices.

HRP has, however, a practical blind spot: it cannot respect constraints. Real mandates impose per-asset limits (UCITS concentration rules, single-stock caps), sector or asset-class caps and floors (investment policy statements), and turnover budgets (to control trading costs). Recursive bisection has no mechanism for any of these, and in our multi-asset universe unconstrained HRP places on average 86% of capital in a single short-term bond fund, breaching a 15% single-fund cap and a 40% asset-class cap at *every* rebalance. The usual fixes are ad hoc: weights are clipped and renormalised, which can cycle without ever reaching the feasible set, or projected onto it with a Euclidean distance that ignores how assets co-move.

This paper proposes **Constraint-Aware Hierarchical Risk Parity (CA-HRP)**, a principled way to keep HRP's risk allocation while satisfying any set of linear constraints exactly. Our contributions are:

1. **Capacity-aware recursive bisection.** We define a capacity interval for every node of the HRP tree and clip each split into it. We prove that the interval is never empty and that per-asset bounds are met exactly, with no solver (Proposition 1). This yields a solver-free variant, CA-HRP-B.

2. **Tracking-error projection.** We map the HRP target to the feasible portfolio with minimum ex-ante tracking error to it. We show the projection is unique, leaves compliant HRP portfolios unchanged, and redistributes excess weight as a covariance-based hedge rather than uniformly (Proposition 2).

3. **A transparent evaluation protocol.** The method design was selected on a validation period only and then frozen; we report a separate held-out test period, all constraint sets were fixed before any backtest was run, and all code and data are public.

4. **Evidence on six universes and in simulation.** On 20 multi-asset ETFs (2008–2026), 43 NIFTY 50 stocks (2011–2026) and 47 US large caps (2006–2026), and on three external-validation universes added after the method was frozen (Indian ETFs, US sector funds and global country funds), CA-HRP satisfies every constraint at every rebalance and at every rebalancing frequency. Where constraints bind, it is the strongest member of the HRP family; where they rarely bind, it matches HRP. In a Monte Carlo study with a known covariance it has the highest true Sharpe ratio of all methods.

5. **A regime-robust variant and two negative results.** We propose a worst-case tracking-error projection over covariance scenarios with a guarantee over their convex hull (Proposition 3), and report honestly that it does not improve on CA-HRP, and that embedding constraints inside the bisection performs worse than projecting afterwards.

The rest of the paper is organised as follows. Section 2 reviews related work, Section 3 states the problem, Section 4 presents the method and its properties, Section 5 describes the data and experimental protocol, Section 6 reports the main results, Section 7 the extended evaluation, and Section 9 concludes.

## 2 Related Work

**Estimation error and robust allocation.** The sensitivity of mean–variance portfolios to input error is well documented [1, 10]. DeMiguel et al. [2] find that across many data sets none of fourteen optimised models consistently beats $1/N$ out of sample. Shrinkage estimators [5] reduce the condition number of the covariance matrix, and Jagannathan and Ma [4] show that no-short-sale constraints behave like covariance shrinkage. Risk-based methods allocate by risk rather than expected return; the equal-risk-contribution (ERC) portfolio [8] equalises marginal risk contributions and can be computed by convex optimisation [16].

**Hierarchical allocation.** López de Prado [7] introduced HRP, combining graph-theoretic clustering with recursive inverse-variance bisection. Raffinot [13] studied hierarchical clustering-based allocation more broadly, and Raffinot [14] proposed the Hierarchical Equal Risk Contribution (HERC) portfolio, which splits along the dendrogram rather than the ordered list. Pfitzinger and Katzke [11] recognised that HRP lacks constraint handling and proposed a cluster-based capital allocation that accommodates weight bounds within the bisection.

**Positioning of this work.** Capacity-aware bisection is closest in spirit to Pfitzinger and Katzke [11]; our contribution is a capacity definition that also incorporates group caps, together with a short proof of non-emptiness and exact box feasibility based on the additivity and sub-additivity of the capacity bounds. More importantly, we show empirically that enforcing constraints *inside* the bisection has a hidden cost: excess weight cascades down one branch and can concentrate in volatile assets. Our main method therefore separates the two concerns: HRP supplies the risk allocation, and a tracking-error projection [15] enforces arbitrary linear constraints (group floors, caps and turnover) with minimum risk distortion. To our knowledge this is the first study to compare constraint-handling strategies for HRP under a common, pre-registered protocol with a held-out test period.

# 3 Problem Setting

We consider $N$ risky assets with daily simple returns $r_t \in \mathbb{R}^N$. At each rebalancing date the manager observes the last $L$ return vectors, estimates a covariance matrix $\hat{\Sigma}$, and chooses weights $w \in \mathbb{R}^N$. The mandate imposes a polyhedral feasible set

$$\mathcal{W}(w^-) = \Big\{ w : \ \mathbf{1}^\top w = 1, \ \ \ell \le w \le u, \ \ \underline{c}_g \le \sum_{i \in g} w_i \le \bar{c}_g \ \ \forall g \in \mathcal{G}, \ \ \|w - w^-\|_1 \le \tau \Big\}, \tag{1}$$

where $\ell, u$ are per-asset bounds, $\mathcal{G}$ partitions the assets into groups (sectors or asset classes) with caps $\bar{c}_g$ and floors $\underline{c}_g$, $w^-$ is the drifted pre-trade portfolio and $\tau$ a turnover budget. All constraints are linear, so $\mathcal{W}$ is a convex polytope. We assume it is non-empty. The turnover budget can in principle conflict with the other constraints if price drift pushes the pre-trade portfolio far outside the box or group limits; our implementation then relaxes the turnover cap for that date only. In all experiments reported here this fallback was never triggered: every CA-HRP portfolio satisfied all constraints, turnover included.

# 4 Method

## 4.1 Hierarchical Risk Parity

HRP [7] proceeds in three steps. (i) *Tree clustering*: the correlation matrix $\rho$ implied by $\hat{\Sigma}$ is mapped to the distance $d_{ij} = \sqrt{(1-\rho_{ij})/2}$ and agglomerated with single linkage. (ii) *Quasi-diagonalisation*: assets are ordered by the dendrogram leaves so that similar assets are adjacent. (iii) *Recursive bisection*: starting from the full ordered list with weight $W = 1$, every cluster $C$ is split into a left and right half $C_L, C_R$; with $V_S = \tilde{w}_S^\top \hat{\Sigma}_{SS} \tilde{w}_S$ the variance of the inverse-variance portfolio $\tilde{w}_S \propto \mathrm{diag}(\hat{\Sigma}_{SS})^{-1}$ of a cluster $S$, the split fraction is

$$\alpha_C = 1 - \frac{V_{C_L}}{V_{C_L} + V_{C_R}}, \qquad W_{C_L} = \alpha_C W_C, \quad W_{C_R} = (1 - \alpha_C) W_C. \tag{2}$$

The recursion ends at single assets. No matrix is inverted and every split is a two-asset risk-parity decision, which is the source of HRP's stability. Nothing in (2), however, knows about $\mathcal{W}$.

### 4.2 Capacity-aware recursive bisection (CA-HRP-B)

Our first contribution makes the bisection itself respect per-asset bounds and, partially, group caps, without any optimisation solver. For a set of assets $S$ define its *capacity interval*

$$\mathrm{lo}(S) = \sum_{i \in S} \ell_i, \qquad \mathrm{hi}(S) = \sum_{g \in \mathcal{G}} \min\Big( \sum_{i \in S \cap g} u_i,\ \bar{c}_g \Big). \tag{3}$$

$\mathrm{lo}(S)$ is the least weight that $S$ must hold and $\mathrm{hi}(S)$ an upper bound on what it can hold. When a cluster $C$ carrying weight $W_C$ is split, the HRP fraction (2) is clipped to

$$\alpha_C \in \Big[ \tfrac{\max(\mathrm{lo}(C_L),\, W_C - \mathrm{hi}(C_R))}{W_C},\quad \tfrac{\min(\mathrm{hi}(C_L),\, W_C - \mathrm{lo}(C_R))}{W_C} \Big]. \tag{4}$$

**Proposition 1** (Exact box feasibility)**.** *If* $\mathrm{lo}(\{1,\dots,N\}) \le 1 \le \mathrm{hi}(\{1,\dots,N\})$*, the interval* (4) *is non-empty at every split, and the resulting weights satisfy* $\mathbf{1}^\top w = 1$ *and* $\ell_i \le w_i \le \min(u_i, \bar{c}_{g(i)})$ *for all* $i$*.*

*Proof.* We show by induction that every cluster satisfies the invariant $\mathrm{lo}(C) \le W_C \le \mathrm{hi}(C)$. It holds at the root by assumption. Note that lo is additive, $\mathrm{lo}(C) = \mathrm{lo}(C_L) + \mathrm{lo}(C_R)$, and hi is sub-additive because $\min(a+b, c) \le \min(a, c) + \min(b, c)$ for $a, b, c \ge 0$, so $\mathrm{hi}(C) \le \mathrm{hi}(C_L) + \mathrm{hi}(C_R)$. The interval (4) (multiplied by $W_C$) is non-empty iff the four inequalities $\mathrm{lo}(C_L) \le \mathrm{hi}(C_L)$, $\mathrm{lo}(C_L) \le W_C - \mathrm{lo}(C_R)$, $W_C - \mathrm{hi}(C_R) \le \mathrm{hi}(C_L)$ and $W_C - \mathrm{hi}(C_R) \le W_C - \mathrm{lo}(C_R)$ hold. The first and last follow from $\ell \le u$; the second from $W_C \ge \mathrm{lo}(C)$ and additivity; the third from $W_C \le \mathrm{hi}(C)$ and sub-additivity. Any $\alpha_C$ in the interval gives children with $\mathrm{lo}(C_L) \le W_{C_L} \le \mathrm{hi}(C_L)$ and likewise for $C_R$, which preserves the invariant. At a leaf $\{i\}$, $\mathrm{lo} = \ell_i$ and $\mathrm{hi} = \min(u_i, \bar{c}_{g(i)})$, which proves the claim. Weights sum to one because each split conserves $W_C$. □

CA-HRP-B costs $O(N^2)$ beyond HRP and needs no solver. It guarantees box bounds exactly, but group constraints are only respected when a group lies inside one sub-tree, and group floors and turnover are not handled. Those remaining constraints are enforced by a light projection (next subsection, with the Euclidean metric replaced by a tree-weighted one; details in the code). As our experiments show, the price of handling constraints *inside* the bisection is that excess weight cascades down a single branch of the tree and can end up concentrated in volatile assets; this motivates the main method.

### 4.3 Tracking-error projection (CA-HRP)

The key observation is that HRP produces an excellent *target* but has no notion of distance between portfolios. Practitioners close the gap by clipping and renormalising, which is not guaranteed to reach $\mathcal{W}$, or by a Euclidean projection $\min_{w \in \mathcal{W}} \|w - w^{\mathrm{HRP}}\|_2^2$, which treats a unit of weight moved from a short-term bond fund to an oil fund the same as one moved to another bond fund. We instead measure distance in *risk*: the ex-ante tracking-error variance of the constrained portfolio relative to HRP [15],

$$w^{\mathrm{CA}} = \arg\min_{w \in \mathcal{W}(w^-)} (w - w^{\mathrm{HRP}})^\top \bar{\Sigma}\, (w - w^{\mathrm{HRP}}), \qquad \bar{\Sigma} = \frac{N \hat{\Sigma}_{\mathrm{LW}}}{\mathrm{tr}\, \hat{\Sigma}_{\mathrm{LW}}} + \epsilon I, \tag{5}$$

where $\hat{\Sigma}_{\mathrm{LW}}$ is the Ledoit–Wolf shrinkage estimator [5], $w^{\mathrm{HRP}}$ is HRP computed on $\hat{\Sigma}_{\mathrm{LW}}$, and the small ridge $\epsilon = 10^{-4}$ guarantees strict convexity.

**Proposition 2** (Properties of the projection)**.** *(a) If* $\mathcal{W}(w^-) \neq \emptyset$*, problem* (5) *has a unique solution. (b) If* $w^{\mathrm{HRP}} \in \mathcal{W}(w^-)$ *then* $w^{\mathrm{CA}} = w^{\mathrm{HRP}}$*: CA-HRP never alters HRP when HRP already complies. (c) Among all feasible portfolios,* $w^{\mathrm{CA}}$ *minimises the ex-ante variance of the return difference to HRP, so it is the feasible portfolio whose returns are most correlated with, and least volatile around, the HRP portfolio. (d) Let* $Aw \le b$ *collect the active inequality constraints*

**Algorithm 1** Constraint-Aware HRP (one rebalancing date)

**Require:** returns window $R \in \mathbb{R}^{L\times N}$, feasible set $\mathcal{W}$, drifted weights $w^-$

1: $\hat{\Sigma}_{\mathrm{LW}} \leftarrow$ Ledoit–Wolf shrinkage covariance of $R$
2: $\rho \leftarrow$ correlation of $\hat{\Sigma}_{\mathrm{LW}}$; $d_{ij} \leftarrow \sqrt{(1-\rho_{ij})/2}$
3: $\pi \leftarrow$ leaf order of the single-linkage dendrogram of $d$
4: $w^{\mathrm{HRP}} \leftarrow$ recursive bisection (2) on $\pi$ ▷ optionally clipped by (4)
5: $w^{\mathrm{CA}} \leftarrow$ solve the convex QP (5) over $\mathcal{W}(w^-)$
6: **return** $w^{\mathrm{CA}}$

*and* $\mathbf{1}^\top w = 1$ *the budget. The KKT conditions give* $w^{\mathrm{CA}} = w^{\mathrm{HRP}} - \frac{1}{2}\bar{\Sigma}^{-1}(A^\top\lambda + \nu\mathbf{1})$ *with* $\lambda \geq 0$*: the correction is a* hedge. *Weight removed from a capped asset is redistributed in proportion to the rows of* $\bar{\Sigma}^{-1}$*, i.e. towards the assets that best replicate the capped asset's risk.*

*Proof.* (a) $\bar{\Sigma} \succ 0$, so the objective is strictly convex and $\mathcal{W}$ is a non-empty closed convex set; a unique minimiser exists. (b) The objective is non-negative and zero at $w^{\mathrm{HRP}}$. (c) $(w - w^{\mathrm{HRP}})^\top\hat{\Sigma}(w - w^{\mathrm{HRP}})$ is the variance of $(w - w^{\mathrm{HRP}})^\top r$ under the estimated covariance; the trace normalisation and ridge only rescale and regularise it. (d) Stationarity of the Lagrangian $(w - w^{\mathrm{HRP}})^\top\bar{\Sigma}(w - w^{\mathrm{HRP}}) + \lambda^\top(Aw - b) + \nu(\mathbf{1}^\top w - 1)$, with the $\ell_1$ turnover constraint written in its standard linear epigraph form. □

Part (d) explains the behaviour we observe in the multi-asset universe: when HRP wants far more than the 40% asset-class cap in bonds, the Euclidean projection spreads the excess almost uniformly (about 5% to each non-bond fund), while the tracking-error projection re-shapes the portfolio as a hedge: inside the bond budget it moves from short-term to intermediate and long Treasuries (IEF, TLT), whose negative correlation with equities offsets the risk of the assets that must now be held, and outside the bond budget it favours gold and broad commodities (Section 6). The constrained portfolio keeps the defensive character of HRP, with markedly lower volatility than the Euclidean alternative.

**Complexity.** Steps 1–4 cost $O(LN^2 + N^2 \log N)$. Step 5 is a QP with $N$ variables (plus $N$ auxiliary variables for the turnover epigraph) and $N + |\mathcal{G}|$ linear constraints; with an interior-point solver [3] it takes a few milliseconds for $N \leq 50$. A full 20-year monthly backtest of CA-HRP runs in under six seconds on a laptop; Section 7 reports timings up to $N = 400$.

### 4.4 Regime-robust projection (CA-HRP-R)

The tracking-error metric in (5) trusts a single covariance estimate. When correlations shift, the hedge chosen by the projection can fail. We therefore also study a *regime-robust* projection that tracks HRP under the worst of $K$ covariance scenarios $\{\Sigma_k\}_{k=1}^K$:

$$w^{\mathrm{R}} = \arg\min_{w\in\mathcal{W}(w^-)} \max_{k=1,\dots,K} (w - w^{\mathrm{HRP}})^\top\bar{\Sigma}_k(w - w^{\mathrm{HRP}}), \tag{6}$$

where every $\bar{\Sigma}_k$ uses the same scale factor as the baseline estimate, so that relative risk levels across regimes are preserved. We consider three scenarios: the 1-year Ledoit–Wolf estimate, a 3-month (63-day) Ledoit–Wolf estimate capturing the current regime, and a "hedges-fail" stress covariance $D|\rho|D$ in which every correlation is replaced by its absolute value.

**Proposition 3** (Robust projection)**.** *Problem* (6) *is convex, has a unique solution, and returns* $w^{\mathrm{HRP}}$ *whenever it is feasible. Its optimal value* $t^\star$ *bounds the ex-ante tracking-error variance under* every *covariance matrix in the convex hull* $\mathrm{conv}\{\bar{\Sigma}_1, \dots, \bar{\Sigma}_K\}$*:* $(w^{\mathrm{R}} - w^{\mathrm{HRP}})^\top\Sigma(w^{\mathrm{R}} - w^{\mathrm{HRP}}) \leq t^\star$.

*Proof.* Each term is a strictly convex quadratic, and the pointwise maximum of strictly convex functions is strictly convex; with a non-empty convex feasible set this gives existence and uniqueness, and the objective is zero only at $w^{\mathrm{HRP}}$. For $\Sigma = \sum_k \lambda_k\bar{\Sigma}_k$ with $\lambda$ in the simplex and

$d = w - w^{\mathrm{HRP}}$, $d^\top \Sigma d = \sum_k \lambda_k d^\top \bar{\Sigma}_k d \le \max_k d^\top \bar{\Sigma}_k d$, because the tracking-error variance is linear in $\Sigma$. □

Problem (6) is a second-order cone program (epigraph form) and solves as fast as (5). The choice of scenarios was made on the validation period only (Appendix A): the variant using the 1-year and 3-month regimes was selected. The hedges-fail stress scenario was rejected at that stage because it lowered validation performance markedly, even though the held-out period later contained exactly such a regime (2022). We report the frozen choice regardless.

# 5 Experimental Design

**Data.** We use daily dividend- and split-adjusted closing prices from Yahoo Finance for three universes chosen to span different constraint regimes (Table 1): (i) 20 US-listed multi-asset ETFs covering equities, government and corporate bonds, commodities and real estate (January 2008 – October 2026); (ii) 43 long-listed constituents of India's NIFTY 50 index across 12 sectors (January 2011 – October 2026); (iii) 47 US large-cap stocks across 11 sectors (January 2006 – October 2026). All series are complete over their sample. Survivorship bias affects the stock universes, which consist of current large caps; it affects all methods equally and does not change their ranking, which is our object of study.

Table 1: Universes and constraint sets. Constraint sets were fixed before any backtest.

| | Multi-asset ETFs | NIFTY 50 stocks | US large caps |
|---|---|---|---|
| Assets / groups | 20 / 4 | 43 / 12 | 47 / 11 |
| Sample | 2008–2026 | 2011–2026 | 2006–2026 |
| Per-asset cap $u_i$ | 15% | 6% | 6% |
| Group cap $\bar{c}_g$ | 40% | 25% | 20% |
| Group floor $\underline{c}_g$ | Equity $\geq$ 20% | – | – |
| Turnover budget $\tau$ (per month) | 30% | 30% | 30% |
| Validation / test split | 2016 \| 2017– | 2018 \| 2019– | 2015 \| 2016– |

**Constraints.** The constraint sets mirror typical mandates: a single-holding cap of roughly $2.5/N$, sector or asset-class caps, a growth floor of 20% in equities for the multi-asset portfolio, and a monthly two-sided turnover budget of 30%. All weights are long-only and fully invested.

**Methods compared.** *Unconstrained references*: equal weight (EW), inverse variance (IVP), long-only minimum variance (MinVar), equal risk contribution (ERC) and standard HRP. *Constrained competitors*: minimum variance (MinVar-C) and mean–variance with risk aversion 5 (MVO-C), both solved over the full feasible set; HRP with iterative clipping and renormalisation (HRP-Clip, the common practitioner fix, which ignores turnover); and HRP followed by Euclidean projection onto the full feasible set (HRP-Proj). *Proposed*: the solver-free capacity-aware bisection (CA-HRP-B) and the full method with tracking-error projection (CA-HRP). Covariance-based competitors use the Ledoit–Wolf estimator, except the HRP baselines, which follow the original sample-covariance recipe.

**Backtest protocol.** At each month end we estimate inputs from the previous $L = 252$ trading days, compute target weights, and trade at the close. Between rebalances weights drift with prices. Turnover is $\|w_t - w_t^-\|_1$ against the drifted weights, and a cost of 10 bps per unit of turnover is deducted (we also report 0, 25 and 50 bps). Constraint compliance is checked on the target weights at every rebalance.

**Design selection and held-out test.** An initial version of CA-HRP enforced all constraints inside the bisection followed by a tree-weighted projection. Before finalising the method, we compared eight configurations *on the validation period only* (first part of each sample, Table 1) and selected the tracking-error projection with Ledoit–Wolf covariance (Appendix A). The configuration was then frozen, and we report results both over the full sample and over the untouched test period.

**Metrics.** Annualised return (CAGR), volatility, Sharpe ratio (zero risk-free rate), maximum drawdown, daily 5% CVaR, annualised turnover, constraint-violation rate (share of rebalances with any breach), realised tracking error to unconstrained HRP ($\mathrm{TE}_{\mathrm{HRP}}$), effective number of holdings $1/\sum_i w_i^2$, and risk concentration, the Herfindahl index of percentage risk contributions scaled by $N$ (1 = equal risk). Sharpe ratio differences are tested with a circular block bootstrap (block of 21 days, 2,000 replications) in the spirit of Ledoit and Wolf [6] and Politis and Romano [12].

# 6 Results

## 6.1 Constraint compliance

Table 2 first answers the practical question. Unconstrained HRP breaches the mandate at 100% of rebalances in the multi-asset universe, 48% for NIFTY and 66% for US stocks. The common practitioner fix, HRP-Clip, still breaches constraints at 46%, 8% and 8% of rebalances respectively: alternating clipping and group rescaling can cycle when per-asset and group limits interact, and it ignores turnover entirely. Every optimisation-based method, including both CA-HRP variants, satisfies all constraints at every rebalance, and CA-HRP never needed the turnover fallback.

## 6.2 Performance where constraints bind

The multi-asset universe is where constraints bite hardest: HRP would hold on average 86% in a single short-term Treasury fund (Figure 2), so the 15% fund cap, the 40% asset-class cap and the 20% equity floor force a very different portfolio. Here the way the constraints are enforced matters a great deal. CA-HRP reaches a Sharpe ratio of 0.908 with 7.4% volatility and a −18.8% maximum drawdown, against 0.764, 9.9% and −22.8% for the Euclidean projection (HRP-Proj) and 0.709 for HRP-Clip. The difference with HRP-Clip is significant ($p = 0.01$); against HRP-Proj and equal weight the $p$-values are 0.13 and 0.12. Among all constrained methods, only constrained minimum variance has a higher Sharpe ratio (0.935), a difference that is far from significant ($p = 0.34$). CA-HRP's realised tracking error to HRP (6.4%) is the lowest of the HRP-family methods, as its construction intends. The composition explains why. Averaged over all rebalances, HRP-Proj spreads the weight it removes from bonds almost evenly (about 5% in each non-bond fund), whereas CA-HRP keeps the bond budget but shifts it from short-term to intermediate and long Treasuries (IEF 13%, TLT 9%), whose negative correlation with equities hedges the risk it must now take, and holds gold (13%), broad commodities (12%) and the S&P 500 (11%) as its main risky positions.

The solver-free CA-HRP-B performs like the Euclidean projection (0.750) but with four times the turnover. Its bisection handles constraints by pushing excess weight down the tree, so the portfolio concentrates in whichever volatile branch has spare capacity (commodities, Figure 2 and Section 4). This is the central negative finding that motivated the final design: constraints are best enforced *after* the risk allocation, in a risk metric, not inside the bisection.

## 6.3 Performance where constraints rarely bind

In the two stock universes HRP breaches the limits at about half of the rebalances, but only by small amounts: the realised tracking error between HRP and any of its constrained versions is at most 1%. All HRP-based methods are therefore statistically indistinguishable, which is what

Table 2: Full-sample out-of-sample results, net of 10 bps costs. CAGR, Vol, MaxDD, CVaR and $TE_{HRP}$ in %; turnover annualised; Viol = share of rebalances breaching any constraint; $p$ = bootstrap $p$-value of the Sharpe difference with CA-HRP. Unconstrained methods (top block) are shown for reference only. Best constrained Sharpe ratio in bold.

| Universe | Method | CAGR | Vol | Sharpe | $p$ | MaxDD | $CVaR_5$ | Turn. | Viol | $TE_{HRP}$ | Eff. N |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Multi-asset ETFs | EW | 8.0 | 10.9 | 0.763 | 0.12 | -24.8 | -1.63 | 0.34 | 0% | 10.2 | 20.0 |
| | IVP | 1.9 | 2.2 | 0.840 | 0.70 | -9.6 | -0.34 | 0.30 | 100% | 0.5 | 1.6 |
| | MinVar | 2.4 | 2.4 | 0.971 | 0.68 | -9.1 | -0.36 | 0.90 | 100% | 1.4 | 2.5 |
| | ERC | 4.8 | 5.7 | 0.852 | 0.47 | -17.3 | -0.84 | 0.48 | 100% | 4.5 | 9.3 |
| | HRP | 1.6 | 2.0 | 0.833 | 0.70 | -8.8 | -0.30 | 0.63 | 100% | 0.0 | 1.4 |
| | MinVar-C | 6.8 | 7.3 | **0.935** | 0.34 | -17.5 | -1.07 | 1.21 | 0% | 6.5 | 8.0 |
| | MVO-C | 8.8 | 12.2 | 0.753 | 0.30 | -19.1 | -1.91 | 3.16 | 0% | 11.6 | 7.3 |
| | HRP-Clip | 6.3 | 9.2 | 0.709 | 0.01 | -22.8 | -1.37 | 2.17 | 46% | 8.4 | 13.9 |
| | HRP-Proj | 7.4 | 9.9 | 0.764 | 0.13 | -22.8 | -1.49 | 0.62 | 0% | 9.3 | 16.1 |
| | CA-HRP-B | 7.1 | 9.8 | 0.750 | 0.14 | -21.4 | -1.47 | 2.50 | 0% | 9.2 | 10.7 |
| | **CA-HRP** | 6.6 | 7.4 | 0.908 | – | -18.8 | -1.08 | 1.25 | 0% | 6.4 | 8.3 |
| NIFTY 50 stocks | EW | 19.7 | 15.4 | 1.247 | 0.07 | -36.0 | -2.21 | 0.62 | 0% | 3.7 | 43.0 |
| | IVP | 18.7 | 14.3 | 1.272 | 0.02 | -33.4 | -2.04 | 0.85 | 17% | 2.0 | 37.8 |
| | MinVar | 18.0 | 12.8 | 1.356 | 0.89 | -26.1 | -1.74 | 2.87 | 100% | 4.7 | 14.1 |
| | ERC | 19.5 | 14.2 | 1.326 | 0.22 | -32.7 | -2.02 | 0.81 | 3% | 1.8 | 39.0 |
| | HRP | 19.5 | 13.7 | 1.368 | 0.96 | -30.6 | -1.94 | 2.70 | 48% | 0.0 | 32.4 |
| | MinVar-C | 18.6 | 12.8 | **1.401** | 0.62 | -26.5 | -1.76 | 2.12 | 0% | 3.5 | 19.8 |
| | MVO-C | 20.2 | 16.2 | 1.216 | 0.17 | -33.2 | -2.35 | 3.35 | 0% | 6.8 | 17.4 |
| | HRP-Clip | 19.6 | 13.7 | 1.369 | 1.00 | -30.6 | -1.95 | 2.66 | 8% | 0.2 | 32.8 |
| | HRP-Proj | 19.6 | 13.8 | 1.368 | 0.97 | -30.6 | -1.95 | 2.62 | 0% | 0.2 | 32.9 |
| | CA-HRP-B | 19.5 | 13.7 | 1.368 | 0.68 | -30.1 | -1.95 | 2.51 | 0% | 0.9 | 33.1 |
| | **CA-HRP** | 19.5 | 13.7 | 1.369 | – | -30.1 | -1.94 | 2.47 | 0% | 0.9 | 33.1 |
| US large caps | EW | 13.3 | 18.5 | 0.767 | 0.59 | -47.9 | -2.80 | 0.53 | 0% | 4.8 | 47.0 |
| | IVP | 11.7 | 16.1 | 0.768 | 0.15 | -41.1 | -2.40 | 0.73 | 24% | 1.3 | 37.0 |
| | MinVar | 9.3 | 13.7 | 0.720 | 0.39 | -33.0 | -1.98 | 3.02 | 100% | 6.2 | 10.5 |
| | ERC | 12.4 | 16.8 | 0.781 | 0.67 | -44.7 | -2.54 | 0.68 | 3% | 2.3 | 42.6 |
| | HRP | 11.6 | 15.7 | 0.776 | 0.14 | -40.7 | -2.35 | 2.88 | 66% | 0.0 | 33.9 |
| | MinVar-C | 10.7 | 14.5 | 0.773 | 0.66 | -35.7 | -2.15 | 2.06 | 0% | 3.6 | 19.5 |
| | MVO-C | 12.7 | 17.5 | 0.774 | 0.82 | -43.2 | -2.64 | 3.43 | 0% | 6.9 | 17.5 |
| | HRP-Clip | 11.6 | 15.8 | 0.775 | 0.12 | -41.1 | -2.37 | 2.80 | 8% | 0.3 | 34.6 |
| | HRP-Proj | 11.7 | 15.8 | 0.775 | 0.11 | -41.2 | -2.37 | 2.76 | 0% | 0.3 | 34.7 |
| | CA-HRP-B | 11.9 | 15.8 | 0.790 | 0.55 | -41.9 | -2.36 | 2.65 | 0% | 1.0 | 34.8 |
| | **CA-HRP** | 11.9 | 15.7 | **0.793** | – | -40.4 | -2.35 | 2.57 | 0% | 0.9 | 34.7 |

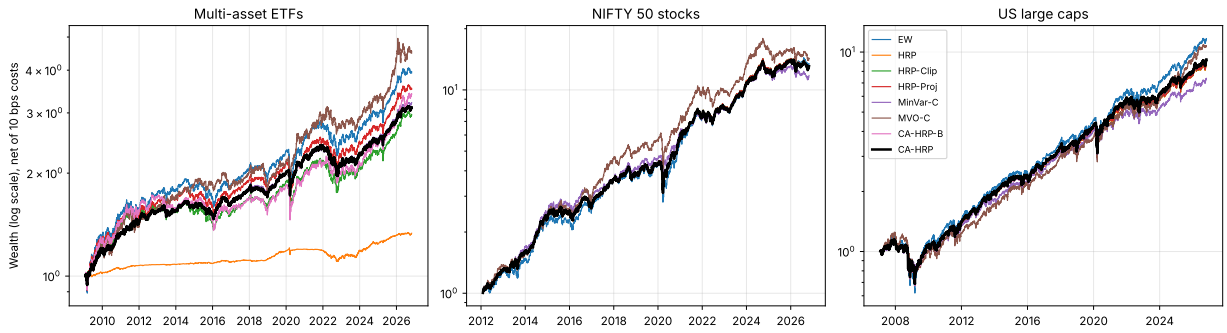



Figure 1: Growth of one unit of wealth (log scale), net of 10 bps costs.

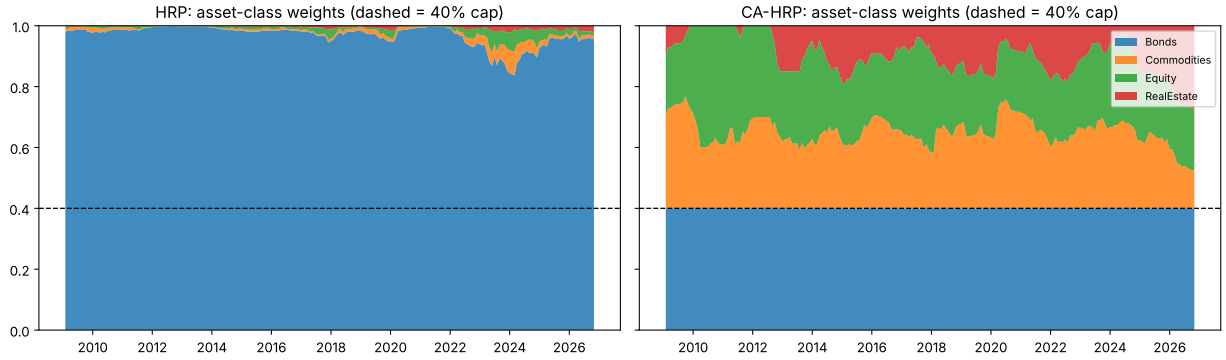



Figure 2: Asset-class weights in the multi-asset universe. Unconstrained HRP (left) is almost entirely in bonds, far above the 40% cap. CA-HRP (right) holds bonds at the cap and allocates the remainder by tracking-error projection.

Proposition 2(b) predicts as the constraints become slack. CA-HRP posts the highest Sharpe ratio of all eleven methods for US large caps (0.793) and ties HRP for NIFTY (1.369 vs 1.368), where constrained minimum variance is marginally higher (1.401, $p = 0.62$). In other words, CA-HRP costs nothing when HRP is already compliant and helps when it is not.

### 6.4 Held-out test period

Table 3 repeats the comparison on the period after the validation window, which played no role in choosing the design. The ranking is unchanged. In the multi-asset universe CA-HRP's test Sharpe is 0.929, against 0.805 for HRP-Proj, 0.802 for HRP-Clip, 0.783 for equal weight and 0.977 for constrained minimum variance, with the lowest volatility of any HRP-family method. In stocks CA-HRP again matches HRP.

### 6.5 Ablation

Table 4 isolates each design choice. In the multi-asset universe the *distance metric* drives almost the whole gain: replacing the full-covariance tracking-error metric by its diagonal lowers the Sharpe ratio from 0.908 to 0.777, and the Euclidean metric to 0.742. Adding capacity-aware bisection before the projection lowers it to 0.861 and nearly doubles turnover; adding a tree-consistency penalty has a similar effect. The choice of covariance estimator, linkage rule and split rule matters little (Sharpe 0.910–0.922), so the method is not sensitive to those HRP details. Removing the turnover cap barely changes performance, showing that the cap is rarely binding for CA-HRP and does not drive its results. In the stock universes no ablation moves the Sharpe ratio by more than 0.04.

Table 3: Sharpe ratio (net of 10 bps) on the held-out test period, which was never used to choose the CA-HRP design, next to the full sample. Vol and MaxDD (%) on the test period.

| | Test-period Sharpe | | | ETFs, test period | | |
|---|---|---|---|---|---|---|
| Method | ETFs | NIFTY | US | Vol | MaxDD | Full-sample SR |
| EW | 0.783 | 1.066 | 0.972 | 10.6 | -24.8 | 0.763 |
| HRP | 0.830 | 1.126 | 0.909 | 2.4 | -8.8 | 0.833 |
| MinVar-C | 0.977 | 1.051 | 0.889 | 7.4 | -17.5 | 0.935 |
| MVO-C | 0.781 | 0.869 | 0.976 | 12.9 | -19.1 | 0.753 |
| HRP-Clip | 0.802 | 1.126 | 0.909 | 8.9 | -22.8 | 0.709 |
| HRP-Proj | 0.805 | 1.126 | 0.910 | 9.6 | -22.8 | 0.764 |
| CA-HRP-B | 0.876 | 1.122 | 0.947 | 9.5 | -21.3 | 0.750 |
| **CA-HRP** | 0.929 | 1.123 | 0.941 | 7.4 | -18.8 | 0.908 |

Table 4: Ablation of CA-HRP components (full sample, net of 10 bps). Each row changes one component of the full method.

| | Sharpe ratio | | | ETFs | | |
|---|---|---|---|---|---|---|
| Variant | ETFs | NIFTY | US | Vol | Turn. | $TE_{HRP}$ |
| CA-HRP (full) | 0.908 | 1.369 | 0.793 | 7.4 | 1.25 | 6.4 |
| sample covariance | 0.922 | 1.372 | 0.781 | 7.3 | 1.22 | 6.4 |
| diagonal metric | 0.777 | 1.366 | 0.791 | 9.2 | 1.14 | 8.5 |
| Euclidean metric | 0.742 | 1.365 | 0.789 | 9.8 | 1.06 | 9.1 |
| + capacity bisection | 0.861 | 1.367 | 0.796 | 8.5 | 2.30 | 7.7 |
| + tree-consistency term | 0.846 | 1.370 | 0.794 | 7.9 | 2.27 | 7.1 |
| no turnover cap | 0.910 | 1.369 | 0.793 | 7.3 | 1.26 | 6.4 |
| Ward linkage | 0.910 | 1.330 | 0.759 | 7.4 | 1.30 | 6.4 |
| dendrogram splits | 0.918 | 1.392 | 0.778 | 7.4 | 1.19 | 6.5 |

### 6.6 Robustness

**Costs and estimation window.** Table 5 shows the multi-asset advantage of CA-HRP over the other HRP-family methods survives 50 bps costs (0.840 vs 0.739 for HRP-Proj) and both shorter (126-day) and longer (504-day) estimation windows, where CA-HRP improves on HRP-Proj by 0.14 and 0.19 Sharpe points respectively. Constrained minimum variance remains a strong competitor throughout.

**Stability of weights.** We re-estimate every method on 30 block-bootstrapped versions of the estimation window at eight dates per universe and measure how much the weights move. CA-HRP is far more stable than mean–variance (instability 0.16–0.22 vs 0.70–0.78) and more stable than constrained minimum variance in every universe (e.g. 0.218 vs 0.324 for NIFTY). In the multi-asset universe HRP-Proj is the most stable, because it is anchored near equal weight; its stability comes at the price of a portfolio that tracks HRP poorly and has higher volatility.

Table 5: Robustness. Left: Sharpe ratio in the multi-asset universe at 50 bps costs and with 126- and 504-day estimation windows (all universes). Right: weight instability, the mean $L_1$ distance of bootstrapped weights from their average (lower = more stable).

| | ETFs | Window 126 days | | | Window 504 days | | | Weight instability | | |
|---|---|---|---|---|---|---|---|---|---|---|
| Method | 50 bps | ETFs | NIFTY | US | ETFs | NIFTY | US | ETFs | NIFTY | US |
| EW | 0.750 | 0.535 | 1.197 | 0.794 | 0.701 | 1.196 | 0.775 | – | – | – |
| HRP | 0.705 | 0.622 | 1.288 | 0.786 | 0.932 | 1.287 | 0.767 | 0.068 | 0.230 | 0.227 |
| MinVar-C | 0.868 | 0.703 | 1.392 | 0.801 | 0.936 | 1.341 | 0.730 | 0.185 | 0.324 | 0.318 |
| MVO-C | 0.650 | 0.660 | 1.142 | 0.755 | 0.796 | 1.172 | 0.760 | 0.754 | 0.698 | 0.781 |
| HRP-Clip | 0.614 | 0.479 | 1.293 | 0.786 | 0.800 | 1.288 | 0.767 | 0.209 | 0.222 | 0.214 |
| HRP-Proj | 0.739 | 0.535 | 1.293 | 0.792 | 0.714 | 1.288 | 0.766 | 0.037 | 0.220 | 0.217 |
| **CA-HRP** | 0.840 | 0.676 | 1.301 | 0.803 | 0.908 | 1.276 | 0.766 | 0.158 | 0.218 | 0.211 |

**Stress periods.** Table 6 reports cumulative returns in three crises. In the COVID crash CA-HRP lost 6.9% in the multi-asset universe, against 14.0% for HRP-Proj, 13.7% for HRP-Clip and 14.9% for equal weight, and it was flat in the GFC window. The 2022 rates shock is the exception: CA-HRP's hedge tilt towards intermediate and long Treasuries, which protected it in 2020, lost 17.2% when bonds and equities fell together, slightly more than HRP-Proj (−15.3%). This illustrates the main limitation of any covariance-based hedge: it relies on correlations that can break down.

Table 6: Cumulative return (%) in stress windows: GFC = Sep 2008–Mar 2009, COVID = 15 Feb–30 Apr 2020, Rates = Jan–Oct 2022.

| | ETFs | | | NIFTY | US | | |
|---|---|---|---|---|---|---|---|
| Method | GFC | COVID | Rates | COVID | GFC | COVID | Rates |
| EW | -1.6 | -14.9 | -16.9 | -16.8 | -30.4 | -13.5 | -5.1 |
| HRP | 0.6 | -0.3 | -6.6 | -9.6 | -24.8 | -10.0 | -5.9 |
| MinVar-C | 0.5 | -7.6 | -16.0 | -4.4 | -20.3 | -10.7 | -4.4 |
| MVO-C | 0.8 | -4.5 | -8.2 | -13.8 | -30.1 | -9.3 | -0.9 |
| HRP-Clip | 0.4 | -13.7 | -17.4 | -9.6 | -24.8 | -10.0 | -6.3 |
| HRP-Proj | -1.0 | -14.0 | -15.3 | -9.6 | -25.0 | -10.0 | -6.3 |
| CA-HRP-B | -2.2 | -11.2 | -11.3 | -8.4 | -25.8 | -9.0 | -5.3 |
| **CA-HRP** | 0.3 | -6.9 | -17.2 | -8.4 | -24.7 | -9.5 | -5.4 |

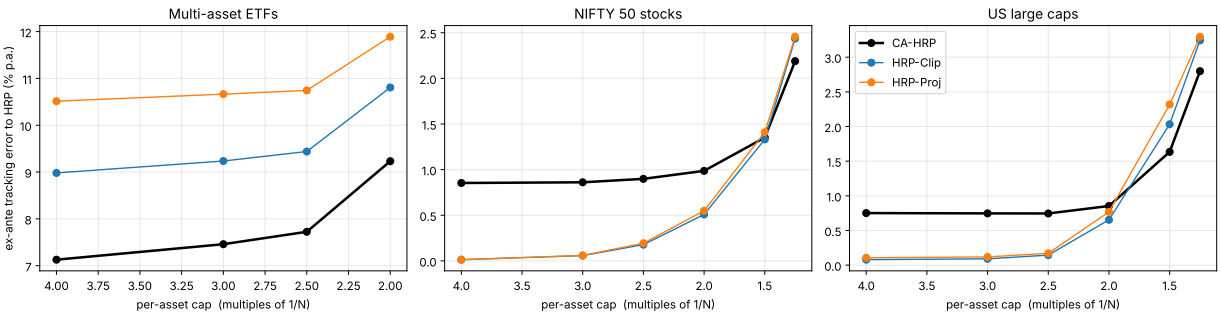



Figure 3: Ex-ante tracking error to sample-covariance HRP as the per-asset cap tightens (right = tighter). Infeasible settings are omitted.

**Tightening the constraints.** Finally, we tighten the per-asset cap from $4/N$ to $1.25/N$ (with group caps scaled accordingly) and measure, at twelve dates per universe, how far each method's portfolio is from HRP in ex-ante tracking-error terms (Figure 3). Settings that are mathematically infeasible (caps of $1.5/N$ and $1.25/N$ in the multi-asset universe) are detected by a linear feasibility check and skipped; HRP-Clip silently returns infeasible weights in such cases, and even at caps of $2.5/N$ and looser it still breaches limits by 1.1–1.9 percentage points on average in the multi-asset universe. In that universe CA-HRP has the smallest distortion at every cap (7.1–9.2% versus 10.5–11.9% for HRP-Proj) and the lowest volatility. In the stock universes CA-HRP has a small constant floor of about 0.8% because its HRP target uses shrinkage rather than the sample covariance; at the tightest cap ($1.25/N$) it distorts HRP the least in both stock universes (2.2% vs 2.4–2.5% for NIFTY, 2.8% vs 3.2–3.3% for US), and already at $1.5/N$ for US stocks.

# 7 Extended Evaluation

After the main study was complete and the method frozen, we ran a second, broader round of tests designed to probe its weaknesses rather than its strengths. All constraint sets and the CA-HRP-R scenario choice were fixed before any of these results were computed.

## 7.1 External validation on three new universes

We added three universes that played no role in any design decision: six Indian exchange-traded funds (Nifty 50, Bank, Junior, PSU Bank, Nasdaq-100 and Gold BeES; April 2011 – October 2026) under a balanced-fund mandate (gold $\leq$ 25%, international equity $\leq$ 35%, Indian equity between 40% and 80%, single fund $\leq$ 35%); the ten US sector SPDR funds (June 2000 – October 2026), grouped as cyclical or defensive with defensive sectors capped at 50% and single sectors at 20%; and nineteen single-country equity funds (October 2004 – October 2026) grouped into four regions, each capped at 40% with single countries at 10%. Yahoo Finance records two short-lived unadjusted unit splits in the Indian funds (December 2019 and June 2021), visible as −90% / +900% return pairs; we detect such pairs automatically and rescale the affected prices by the implied split ratio (10 or 100). No other universe is affected.

Table 7 reports the results. Unconstrained HRP would breach these mandates at 94%, 92% and 97% of rebalances respectively, and HRP-Clip still breaches them at 1%, 14% and 9%. CA-HRP is fully compliant. Its Sharpe ratio is above HRP-Proj in the Indian (1.276 vs 1.257) and sector (0.628 vs 0.619) universes and slightly below it for countries (0.428 vs 0.436), and in every universe it has the lowest volatility of the three HRP fixes. Constrained minimum variance (Indian ETFs) and mean–variance (sectors) post the best Sharpe ratios. The external evidence therefore confirms the pattern of the main study: CA-HRP is a reliable way to make HRP compliant, with gains that scale with how hard the constraints bind, but it does not dominate optimisation-based methods.

Table 7: External validation on three universes that were added after the method was frozen (net of 10 bps). Vol and MaxDD in %; Viol = share of rebalances breaching a constraint. Best constrained Sharpe ratio per universe in bold.

| | India ETFs | | | | US sectors | | | | Countries | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Method | Sharpe | Vol | MaxDD | Viol | Sharpe | Vol | MaxDD | Viol | Sharpe | Vol | MaxDD | Viol |
| EW | 1.086 | 13.7 | -31.1 | 0% | 0.586 | 18.3 | -55.2 | 0% | 0.432 | 22.8 | -61.7 | 100% |
| HRP | 1.336 | 10.5 | -21.5 | 94% | 0.641 | 15.7 | -45.9 | 92% | 0.427 | 21.1 | -59.5 | 97% |
| MinVar-C | **1.321** | 10.9 | -24.9 | 0% | 0.653 | 15.7 | -48.1 | 0% | 0.435 | 20.1 | -60.8 | 0% |
| MVO-C | 1.052 | 14.5 | -28.1 | 0% | **0.658** | 17.0 | -47.8 | 0% | 0.386 | 22.1 | -64.8 | 0% |
| HRP-Clip | 1.269 | 11.8 | -25.1 | 1% | 0.613 | 16.6 | -49.1 | 14% | 0.435 | 21.5 | -60.4 | 9% |
| HRP-Proj | 1.257 | 11.9 | -25.6 | 0% | 0.619 | 16.5 | -48.6 | 0% | **0.436** | 21.5 | -60.1 | 0% |
| CA-HRP-B | 1.255 | 11.9 | -25.2 | 0% | 0.603 | 16.7 | -48.9 | 0% | 0.412 | 21.6 | -61.0 | 0% |
| **CA-HRP** | 1.276 | 11.3 | -24.9 | 0% | 0.628 | 16.2 | -47.6 | 0% | 0.428 | 21.2 | -60.0 | 0% |
| CA-HRP-R | 1.275 | 11.3 | -25.6 | 0% | 0.625 | 16.2 | -47.4 | 0% | 0.426 | 21.2 | -60.1 | 0% |

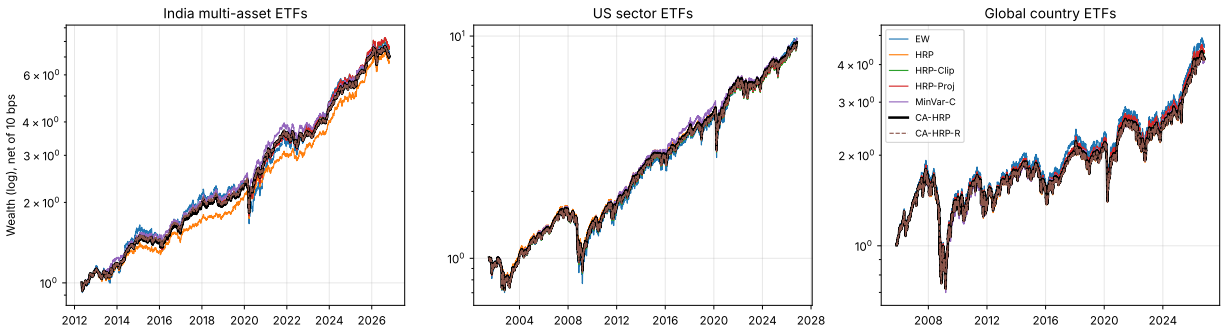



Figure 4: Growth of one unit of wealth (log scale), net of 10 bps, in the three external universes.

## 7.2 Statistical precision across six universes

Table 8 and Figure 5 give bootstrap confidence intervals for the Sharpe-ratio difference between CA-HRP and its competitors. Only one interval excludes zero: CA-HRP versus HRP-Clip in the multi-asset ETF universe (+0.20, 95% interval $[+0.03, +0.35]$). The point estimate against HRP-Proj is positive in five of six universes and is large only for multi-asset ETFs (+0.145, $[-0.04, +0.32]$). Against constrained minimum variance it is negative in five of six universes but never significant. We therefore claim reliability and lower risk, not superior returns.

Table 8: Annualised Sharpe-ratio difference of CA-HRP minus each competitor, with 95% circular-block-bootstrap intervals (21-day blocks, 2,000 draws). Intervals excluding zero in bold.

| Universe | HRP-Proj | HRP-Clip | MinVar-C | EW | CA-HRP-R |
|---|---|---|---|---|---|
| ETFs | +0.145 [-0.04, +0.32] | **+0.199 [+0.03, +0.35]** | -0.027 [-0.08, +0.03] | +0.146 [-0.04, +0.31] | +0.025 [-0.03, +0.08] |
| NIFTY | +0.001 [-0.03, +0.03] | +0.000 [-0.03, +0.03] | -0.032 [-0.15, +0.09] | +0.122 [-0.01, +0.25] | +0.002 [-0.00, +0.00] |
| US | +0.018 [-0.00, +0.04] | +0.018 [-0.00, +0.04] | +0.021 [-0.07, +0.11] | +0.027 [-0.08, +0.12] | +0.000 [-0.00, +0.00] |
| India ETFs | +0.019 [-0.07, +0.11] | +0.007 [-0.06, +0.07] | -0.045 [-0.14, +0.04] | +0.189 [-0.01, +0.38] | +0.001 [-0.03, +0.03] |
| US sectors | +0.010 [-0.01, +0.03] | +0.016 [-0.01, +0.04] | -0.024 [-0.08, +0.02] | +0.043 [-0.02, +0.11] | +0.003 [-0.00, +0.01] |
| Countries | -0.008 [-0.02, +0.01] | -0.007 [-0.02, +0.01] | -0.007 [-0.05, +0.03] | -0.004 [-0.04, +0.03] | +0.002 [-0.00, +0.01] |

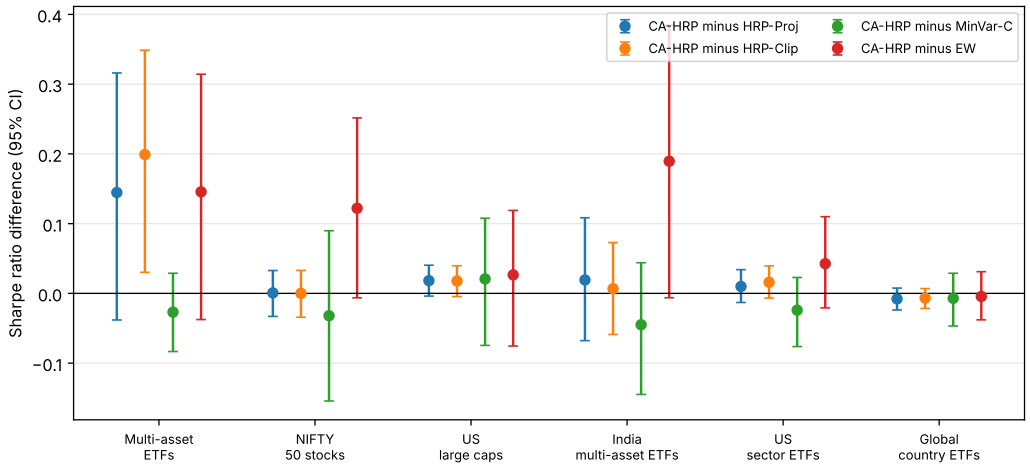



Figure 5: Sharpe-ratio difference of CA-HRP minus each competitor, with 95% block-bootstrap intervals, for all six universes.

## 7.3 Does regime-robustness help?

CA-HRP-R (Section 4.4) was designed to address the 2022 weakness. It did not help. Its Sharpe ratio is lower than CA-HRP's in all six universes, though never by more than 0.025, and it trades more (2.09 vs 1.25 annual turnover in the multi-asset universe). In the 2022 rates shock it lost 18.1% in the multi-asset universe against 17.2% for CA-HRP. The reason is visible in the validation study (Table 13): the only scenario that would have anticipated 2022, the hedges-fail stress covariance, made the portfolio much more conservative in normal times and was rejected on validation data. Robustness to a regime that the estimation window has never seen cannot be bought cheaply with covariance scenarios alone; we report this as a negative result.

## 7.4 Rebalancing frequency

Table 9 repeats the comparison with weekly and quarterly rebalancing (turnover cap of 30% per rebalance in both cases). CA-HRP satisfies all constraints at all frequencies, while HRP-Clip breaches them at 9% (weekly), 14% (monthly) and 27% (quarterly) of rebalances on average

across universes. Over the 18 frequency–universe combinations, CA-HRP beats HRP-Proj in 12 and is the best HRP-based method in 10, while constrained minimum variance beats CA-HRP in 16. The multi-asset ranking (CA-HRP clearly ahead of the other HRP fixes, slightly behind minimum variance) holds at every frequency.

Table 9: Sharpe ratio by rebalancing frequency (W = weekly, M = monthly, Q = quarterly), net of 10 bps, all six universes. The first column also gives HRP-Clip's share of rebalances with a constraint breach, pooled over universes (CA-HRP: 0% at every frequency).

| Freq. | Universe | EW | MinVar-C | HRP-Clip | HRP-Proj | CA-HRP | CA-HRP-R |
|---|---|---|---|---|---|---|---|
| W (Clip 9%) | ETFs | 0.714 | 0.888 | 0.608 | 0.719 | **0.844** | 0.819 |
| | NIFTY | 1.268 | 1.414 | **1.329** | 1.328 | 1.326 | 1.325 |
| | US | 0.779 | 0.781 | **0.765** | 0.763 | 0.763 | 0.763 |
| | India ETFs | 1.066 | 1.331 | 1.258 | 1.241 | **1.269** | 1.261 |
| | US sectors | 0.588 | 0.655 | 0.600 | 0.609 | **0.612** | 0.609 |
| | Countries | 0.434 | 0.438 | 0.404 | **0.407** | 0.397 | 0.396 |
| M (Clip 14%) | ETFs | 0.763 | 0.935 | 0.709 | 0.764 | **0.908** | 0.884 |
| | NIFTY | 1.247 | 1.401 | 1.369 | 1.368 | **1.369** | 1.367 |
| | US | 0.767 | 0.773 | 0.775 | 0.775 | **0.793** | 0.793 |
| | India ETFs | 1.086 | 1.321 | 1.269 | 1.257 | **1.276** | 1.275 |
| | US sectors | 0.586 | 0.653 | 0.613 | 0.619 | **0.628** | 0.625 |
| | Countries | 0.432 | 0.435 | 0.435 | **0.436** | 0.428 | 0.426 |
| Q (Clip 27%) | ETFs | 0.807 | 0.920 | 0.719 | 0.800 | **0.899** | 0.873 |
| | NIFTY | 1.233 | 1.375 | 1.357 | 1.354 | 1.361 | **1.361** |
| | US | 0.787 | 0.744 | **0.812** | 0.808 | 0.807 | 0.808 |
| | India ETFs | 1.107 | 1.317 | 1.279 | 1.261 | 1.295 | **1.299** |
| | US sectors | 0.593 | 0.643 | 0.620 | 0.625 | **0.634** | 0.631 |
| | Countries | 0.415 | 0.416 | 0.421 | **0.423** | 0.409 | 0.409 |

### 7.5 Which constraints matter?

Table 10 imposes the constraint families one at a time. In the multi-asset universe almost all of CA-HRP's advantage comes from the *group* constraints: with group caps and floors alone its Sharpe ratio is 0.972 against 0.745 for HRP-Proj and 0.684 for HRP-Clip, whereas with per-asset caps alone the gap is much smaller (0.853 vs 0.802). This is consistent with Proposition 2(d): group caps remove a large block of correlated weight at once, which is exactly where choosing replacements by covariance rather than uniformly pays off. With per-asset caps alone in the Indian ETF universe, CA-HRP is slightly worse than the simple fixes (1.300 vs 1.312–1.314). HRP-Clip never breaches per-asset caps on their own; its breaches come from group caps and floors (36% of rebalances with groups alone in the multi-asset universe) and, in the sector universe, only from the turnover budget that it ignores.

### 7.6 Monte Carlo evidence with a known covariance

Backtests mix estimation error with luck. To isolate the former we simulate 40 assets in five correlated clusters, one of which is a low-volatility, negatively correlated "bond" block, so that HRP on the *true* covariance puts 90% in that block and breaches a per-asset cap of $2.5/N$ and group caps of 30%. Groups coincide with clusters except for two assets per cluster, so constraints cut across the tree. Every asset has an annualised Sharpe ratio of 0.4. For each of 200 replications per cell we draw $T \in \{126, 252, 504\}$ daily returns (Gaussian or Student-$t_5$), estimate each method, and evaluate the resulting portfolio under the true parameters.

Table 10: Which constraints matter? Sharpe ratio (net of 10 bps) when only some constraint families are imposed. Best HRP-family method per row in bold.

| Universe | Constraints | HRP-Clip | HRP-Proj | CA-HRP |
|---|---|---|---|---|
| ETFs | box only | 0.777 | 0.802 | **0.853** |
| | groups only | 0.684 | 0.745 | **0.972** |
| | box + groups | 0.709 | 0.764 | **0.910** |
| | all (+ turnover) | 0.709 | 0.764 | **0.908** |
| India ETFs | box only | **1.314** | 1.312 | 1.300 |
| | groups only | 1.269 | 1.257 | **1.273** |
| | box + groups | 1.269 | 1.257 | **1.273** |
| | all (+ turnover) | 1.269 | 1.257 | **1.276** |
| US sectors | box only | 0.614 | 0.612 | **0.628** |
| | groups only | 0.626 | 0.635 | **0.639** |
| | box + groups | 0.613 | 0.620 | **0.628** |
| | all (+ turnover) | 0.613 | 0.619 | **0.628** |
| US | box only | 0.775 | 0.774 | **0.792** |
| | groups only | 0.777 | 0.776 | **0.795** |
| | box + groups | 0.775 | 0.774 | **0.793** |
| | all (+ turnover) | 0.775 | 0.775 | **0.793** |

Table 11 and Figure 6 show the Student-$t$ case (the Gaussian case is almost identical). CA-HRP has the highest true Sharpe ratio of all methods in every cell (0.809–0.820 under fat tails, 0.813–0.822 under normality). Among the HRP-based methods it has the lowest true tracking error to the oracle HRP portfolio (6.0–6.3%, against 7.3–7.4% for HRP-Clip and 8.2% for HRP-Proj), on par with constrained minimum variance (within 0.05 percentage points). Its true volatility also matches constrained minimum variance (6.4–6.7%), while HRP-Proj is 8.5% and constrained mean–variance 13.7–14.3%. The cross-replication standard deviation of the true Sharpe ratio is below 0.01, so these differences are precisely estimated. In this controlled setting, which isolates the constraint-handling problem the paper is about, CA-HRP is the best method.

Table 11: Monte Carlo study with a known covariance (40 assets, 5 clusters, 200 replications per cell). True Sharpe ratio and true volatility (%) of the estimated portfolio, and true tracking error (%) to HRP computed on the true covariance. Best value per column among constrained methods in bold.

| | True Sharpe ratio | | | True volatility | | | True TE to true HRP | | |
|---|---|---|---|---|---|---|---|---|---|
| Method (Student-$t_5$) | $T$=126 | 252 | 504 | 126 | 252 | 504 | 126 | 252 | 504 |
| EW | 0.722 | 0.722 | 0.722 | 12.62 | 12.62 | 12.62 | 12.49 | 12.49 | 12.49 |
| MVO-C | 0.680 | 0.683 | 0.684 | 14.00 | 13.80 | 13.86 | 13.85 | 13.65 | 13.71 |
| MinVar-C | 0.808 | 0.809 | 0.811 | **6.63** | **6.52** | 6.44 | **6.28** | 6.17 | 6.09 |
| HRP-Clip | 0.806 | 0.806 | 0.808 | 7.74 | 7.74 | 7.72 | 7.39 | 7.39 | 7.37 |
| HRP-Proj | 0.804 | 0.804 | 0.804 | 8.53 | 8.54 | 8.52 | 8.18 | 8.20 | 8.18 |
| **CA-HRP** | **0.809** | **0.813** | **0.820** | 6.66 | 6.52 | **6.43** | 6.32 | **6.17** | **6.04** |
| CA-HRP-R | 0.808 | 0.808 | 0.811 | 6.73 | 6.64 | 6.57 | 6.39 | 6.31 | 6.22 |

### 7.7 Computational cost

Figure 7 reports the median time per rebalance for random universes of $N = 10$ to 400 assets with ten groups and a turnover cap. CA-HRP takes 17 ms at $N = 50$ and 0.27 s at $N = 400$, the

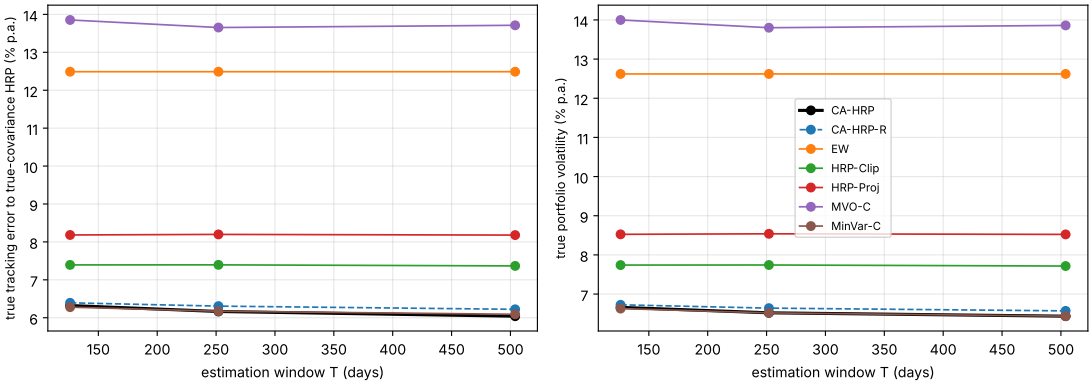



Figure 6: Monte Carlo (Student-$t_5$): true tracking error to the oracle HRP portfolio and true volatility as a function of the estimation window.

same order as constrained minimum variance (0.25 s). The solver-free CA-HRP-B takes 0.15 s at $N = 400$ and the regime-robust CA-HRP-R 1.4 s, so the robust variant is the only one whose cost grows noticeably. All are negligible for monthly rebalancing.

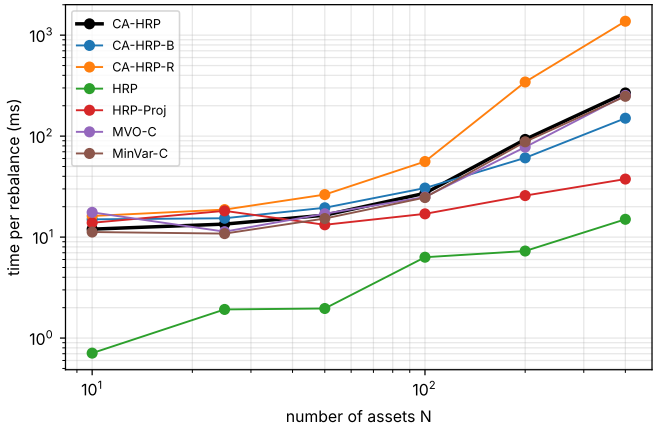



Figure 7: Median computation time per rebalance against the number of assets (log–log).

## 8 Discussion

Four lessons emerge. First, *HRP is not deployable as-is* under realistic mandates: it breached the constraints at 48–100% of rebalances in all six universes, and the clipping fix used in practice still failed at up to 46% (27% on average with quarterly rebalancing). Second, *how* constraints are enforced matters as much as *whether* they are: the same HRP target, projected with a risk-aware metric instead of a Euclidean one, gives a portfolio with 25% less volatility and a higher Sharpe ratio where constraints bind, lower or equal volatility in every universe, and the best true Sharpe ratio in a controlled simulation. The gain comes mostly from group constraints, where whole blocks of correlated weight must be replaced. Third, building constraints into the bisection, while elegant and provably feasible, is inferior to separating risk allocation from constraint enforcement. Fourth, neither the in-bisection approach nor a scenario-robust projection fixed the regime-change weakness; constrained minimum variance remains the method to beat on realised returns.

**Limitations.** The stock universes are survivorship-biased because they contain current large caps; this affects all methods equally. The gains of CA-HRP are concentrated in settings where

constraints bind strongly; when they do not, it reduces to HRP, which is by design. Most Sharpe differences are not statistically significant at conventional levels, which is typical for portfolio comparisons over 15–20 years, and constrained minimum variance remains a strong benchmark. The tracking-error projection inherits the weakness of any covariance-based hedge, visible in 2022 when stock–bond correlation turned positive, and our regime-robust extension did not remove it. All data come from a single public source (Yahoo Finance), which needed one automated correction for unadjusted splits.

# 9 Conclusion

We showed that Hierarchical Risk Parity, despite its robustness, cannot be used directly under realistic portfolio mandates, and that the ad hoc fixes used in practice either fail to restore feasibility or distort the risk allocation. Constraint-Aware HRP separates the two problems: HRP decides how risk should be allocated, and a tracking-error projection finds the compliant portfolio closest to that allocation in risk terms. The method has clean theoretical properties, runs in milliseconds per rebalance, satisfied every constraint in more than 1,300 monthly rebalances across six markets (and at weekly and quarterly frequencies), improved risk-adjusted returns where constraints bind, including on a held-out test period, and was the best method in a controlled simulation. Evidence from three universes added after the design was frozen confirms that its gains scale with how hard the constraints bind. We also report two negative results that we believe are useful for the literature on hierarchical allocation: embedding constraints directly in the recursive bisection, although provably feasible, performs worse than projecting afterwards; and robustness to correlation regimes the estimation window has never seen cannot be obtained simply by adding covariance scenarios.

Future work includes projection metrics that learn regime changes in correlations, the use of CA-HRP with factor-model covariance estimates for universes of thousands of assets, and extending the capacity analysis to non-linear constraints such as tracking-error or CVaR budgets.

**Code and data availability.** All code, data and scripts that reproduce every table and figure are available at `https://github.com/pramukhjain123/capstone`.

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# A Design study on the validation period

Table 12 lists the eight configurations compared before the method was frozen, using only data up to the end of the validation window. The tracking-error projection was best in the multi-asset universe and tied for best in the stock universes. The sample-covariance version scored marginally higher than the Ledoit–Wolf one; we kept Ledoit–Wolf, chosen beforehand for its better conditioning, since the difference is within noise. The initial design ("cap+tree (v0)") was the weakest in the multi-asset universe.

Table 12: Design study on the validation period only (net of 10 bps). "cap" = capacity-aware bisection; "tree" = tree-weighted projection; "TE" = tracking-error projection. LW covariance unless stated.

| | Validation Sharpe | | | ETFs | |
|---|---|---|---|---|---|
| Variant | ETFs | NIFTY | US | Vol | Turn. |
| cap+tree (v0) | 0.605 | 1.737 | 0.639 | 10.2 | 2.59 |
| euclid (LW) | 0.675 | 1.731 | 0.645 | 10.2 | 1.07 |
| cap+euclid | 0.637 | 1.737 | 0.646 | 10.3 | 2.11 |
| diag-var proj | 0.743 | 1.731 | 0.648 | 9.5 | 1.13 |
| TE proj | 0.878 | 1.738 | 0.651 | 7.3 | 1.19 |
| cap+TE proj | 0.750 | 1.737 | 0.651 | 8.6 | 2.45 |
| TE+tree proj | 0.749 | 1.741 | 0.651 | 7.9 | 2.30 |
| TE proj (sample cov) | 0.887 | 1.745 | 0.655 | 7.3 | 1.11 |

Table 13 shows the corresponding study for the regime-robust variant, run later on the same validation windows. A first run of this study contained a bug: the hedges-fail stress matrix was not positive semi-definite, so the solver failed and returned the infeasible HRP target. We repaired the matrix (eigenvalue clipping), added a feasible fallback and a unit test, and reran the study; the selection was unchanged.

Table 13: Validation-period selection of the CA-HRP-R scenario set (Sharpe, net of 10 bps). The robust variant with the best mean was adopted before any test-period or external-universe result was computed.

| Variant | ETFs | NIFTY | US | Mean |
|---|---|---|---|---|
| CA-HRP (reference) | 0.878 | 1.738 | 0.651 | 1.089 |
| max{long,short,stress} | 0.806 | 1.736 | 0.649 | 1.063 |
| max{long,short} | 0.865 | 1.736 | 0.649 | 1.083 |
| max{long,stress} | 0.768 | 1.738 | 0.651 | 1.052 |
| mean{long,short,stress} | 0.779 | 1.737 | 0.650 | 1.055 |

# B Turn-of-month returns in the NIFTY universe

Daily returns on the last trading day of the month are unusually large in the NIFTY sample (an average of 0.42% for an equal-weight portfolio up to 2018, against 0.07% on other days). All methods trade at the month-end close and earn that day's return with their pre-trade weights, so the effect raises all NIFTY Sharpe ratios similarly and does not change their ranking.